

Reference:

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Physics 116A, Mathematical Methods in Physics I

"The Complex Logarithm, Exponential and Power Functions".

Notation: A, B are sets of numbers,

$$\text{sumset } A + B = \{a + b : a \in A, b \in B\}.$$

$$\{0, 1\} + \{2, 3\} = \{2, 3, 4\}.$$

I. Magnitude and Phase

$$z \in \mathbb{C} \Rightarrow z = r \cdot e^{j\theta}.$$

$$r = |z|, \quad \theta = \arg(z).$$

$\Rightarrow \theta$ can take multiple values for a given z .

But, there is a principal value of θ .

fix this principal value, such as

$$\text{Arg}(z) \in [-\pi, \pi)$$

$$\therefore \theta = \text{Arg}(z) + 2\pi k, \quad k \in \mathbb{Z}$$

Denote the set of all possible phases as

$$\arg(z) = \{ \text{Arg}(z) + 2\pi k : k \in \mathbb{Z} \}.$$

Properties:

$$(1) \arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$

$$\underline{\text{LHS:}} \arg(z_1 z_2) = \{ \text{Arg}(z_1) + \text{Arg}(z_2) + 2\pi k : k \in \mathbb{Z} \}.$$

$$\underline{\text{RHS:}} \{ \text{Arg}(z_1) + 2\pi l : l \in \mathbb{Z} \} + \{ \text{Arg}(z_2) + 2\pi n : n \in \mathbb{Z} \}$$

$$(2) \arg(1/z) = -\arg(z)$$

$$= \{ -\text{Arg}(z) + 2\pi k : k \in \mathbb{Z} \}.$$

$$(3) \arg(z_1/z_2) = \arg(z_1 \times \frac{1}{z_2})$$

$$= \arg(z_1) + \arg(1/z_2)$$

$$= \arg(z_1) - \arg(z_2).$$

Warning:

$$\arg(z^2) = \arg(z) + \arg(z)$$

$$\neq 2\arg(z)$$

$$\{ 2\text{Arg}(z) + 2\pi k \}$$

$$2\{ \text{Arg}(z) + 2\pi k \}$$

$$= \{ 2\text{Arg}(z) + 4\pi k \}$$

II. Complex Exponential.

Defn: For any $z \in \mathbb{C}$, define

$$e^z = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$$

Note that:

$$\left. \begin{aligned} \cos(x) &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \\ \sin(x) &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \end{aligned} \right\} x \in \mathbb{R}.$$

$$e^{ix} = \cos x + i \sin x \quad \text{for all } x \in \mathbb{R}.$$

follows from above three equations.

Now, suppose $z_1, z_2 \in \mathbb{C}$

$$e^{(z_1+z_2)} = 1 + \frac{(z_1+z_2)}{1!} + \frac{(z_1+z_2)^2}{2!} + \dots + \frac{(z_1+z_2)^n}{n!} + \dots$$

$(z_1+z_2)^n \rightarrow$ apply binomial expansion

$$\rightarrow \sum_k nC_k \cdot z_1^k z_2^{n-k}$$

Using this idea: $\left(1 + \frac{z_1}{1!} + \frac{z_1^2}{2!} + \dots\right)$

$$\times \left(1 + \frac{z_2}{1!} + \frac{z_2^2}{2!} + \dots\right) = e^{z_1} \cdot e^{z_2}.$$

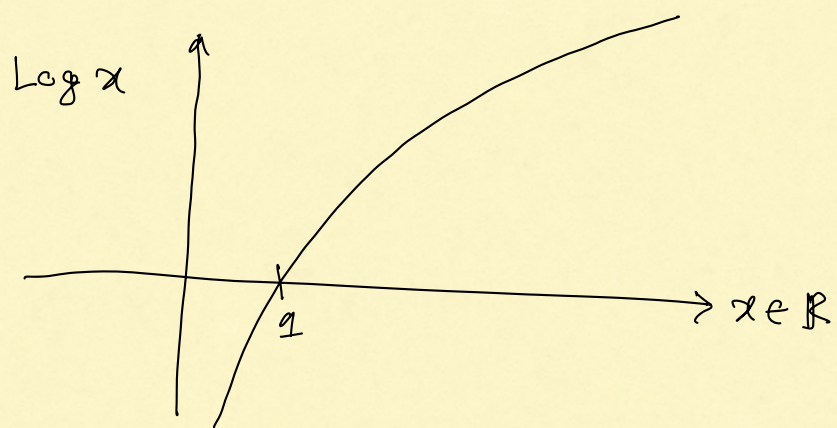
$$\therefore e^{z_1+z_2} = e^{z_1} \cdot e^{z_2}.$$

In particular: if $z = x + iy$, $x, y \in \mathbb{R}$

$$\begin{aligned} e^z &= e^{x+iy} = e^x \cdot e^{iy} \\ &= e^x [\cos y + i \sin y]. \end{aligned}$$

III. Complex Logarithm

Notation: We will denote the usual logarithm (over reals) as Log .



Defn: For any complex $z \neq 0$ we say that w is a logarithm of z if

$$z = e^w = 1 + \frac{w}{1!} + \frac{w^2}{2!} + \dots$$

In general, there are many solutions w .

Suppose $w = u + iv$.

$$z = e^w = e^u \cdot e^{iv}$$

$$\downarrow$$
$$|z| \cdot e^{i \arg(z)} = e^u \cdot e^{iv}$$

$$e^u = |z| \quad \text{and} \quad v \in \arg(z)$$

$$\Rightarrow \begin{matrix} \updownarrow \\ u = \operatorname{Log}|z| \end{matrix} \quad \& \quad v \in \arg(z) \Rightarrow v = \operatorname{Arg}(z) + 2\pi k \quad k \in \mathbb{Z}$$

$$\Rightarrow \log(z) = \underbrace{\operatorname{Log}|z| + i\operatorname{Arg}(z)}_{\substack{\downarrow \\ \text{denote as } \operatorname{Log}(z) \text{ (principal value)}}} + i2\pi k, \quad k \in \mathbb{Z}$$

$$\text{In short } \log(z) = \{ \operatorname{Log}(z) + i2\pi k : k \in \mathbb{Z} \}.$$

Properties:

$$\textcircled{1} \quad \log(e^z) \stackrel{?}{=} z$$

LHS is a set or a multi-valued function

RHS is a complex number. $\Rightarrow \underline{\underline{\log e^z \neq z}}$.

But $z \in \log(e^z)$.

$$\textcircled{2} \quad \log(z_1 z_2) = \log(z_1) + \log(z_2).$$

$$\text{LHS: } \log(|z_1| \cdot e^{j\theta_1} \cdot |z_2| \cdot e^{j\theta_2}).$$

$$= \log(|z_1| \cdot |z_2| \cdot e^{j(\theta_1 + \theta_2)})$$

$$= \operatorname{Log}|z_1| + \operatorname{Log}|z_2| + j\operatorname{Arg}(z_1) + j\operatorname{Arg}(z_2) + j2\pi k, \quad k \in \mathbb{Z}$$

$$= \{ \operatorname{Log}|z_1| + \operatorname{Arg} z_1 + 2\pi l \} + \{ \operatorname{Log}|z_2| + \operatorname{Arg} z_2 + 2\pi m \}$$

$$\Rightarrow \log(z_1 z_2) = \log z_1 + \log z_2$$

$$(3) \log(1/z) = -\log z$$

$$(4) \log(z_1/z_2) = \log z_1 - \log z_2.$$

(5) Warning:

$$\begin{aligned} \log(z^2) &= \log z + \log z \\ &= \{ 2 \log |z| + 2j \operatorname{Arg}(z) + j2\pi k : k \in \mathbb{Z} \} \\ &\neq 2 \log z \end{aligned}$$

$$2 \log z = \{ 2 \log |z| + 2j \operatorname{Arg}(z) + j4\pi k : k \in \mathbb{Z} \}.$$

In general: $\log(z^p) \neq p \log(z).$

However: If p is $\frac{1}{n}$, $n \in \mathbb{Z} \setminus \{0\}$.

$$\boxed{\log(z^{1/n}) = \frac{1}{n} \log z.} \text{ if } n \in \mathbb{Z}, n \neq 0.$$

$$6. e^{\log z} = z.$$

$$\begin{aligned} e^{\log z} &= \exp[\log z + j2\pi k] \\ &= e^{\log z} \cdot e^{j2\pi k}, k \in \mathbb{Z} \\ &= z \cdot e^{j2\pi k}, k \in \mathbb{Z} \\ &= z. \end{aligned}$$

Summary.

$$e^{\log z} = z$$

$$\log(ez) \neq z$$

$$\log(z_1 z_2) = \log z_1 + \log z_2$$

$$\log(z_1 / z_2) = \log z_1 - \log z_2$$

$$\log(z^p) = p \log z \quad \text{if and only if } \frac{1}{p} \text{ is an integer}$$

IV. Complex Powers.

For $z, c \in \mathbb{C}$, what is z^c ?

Defn: z^c is defined as $e^{c \log z}$.

Examples:

$$(1) z = c = i.$$

$$\log z = \log(i) = \log(e^{i\pi/2})$$

$$= \log(1 \cdot e^{i\pi/2})$$

$$= \text{Log} 1 + i\pi/2 + i2\pi k$$

$$= \{i\pi/2 + i2\pi k : k \in \mathbb{Z}\}.$$

$$c \log z = i \{i\pi/2 + i2\pi k\}$$

$$= \{-\pi/2 - 2\pi k\}$$

$$= \{-\pi/2 + 2\pi k : k \in \mathbb{Z}\}.$$

$$e^{c \log z} = \{e^{-\pi/2} \cdot e^{2\pi k} : k \in \mathbb{Z}\}.$$

$$(2) \quad z=1, \quad c=i$$

$$c \log z$$

$$\log z = \log(1 \cdot e^{i0})$$

$$= \log 1 + i0 + i2\pi k$$

$$= \{i2\pi k, k \in \mathbb{Z}\}$$

$$c \log z = \{2\pi k, k \in \mathbb{Z}\}$$

$$i = e^{c \log z} = \{e^{2\pi k}, k \in \mathbb{Z}\}$$

(3) What if c is an integer? $c = k$ an integer.

That is, $z^k = ? \quad k \in \mathbb{Z}$.

$$z^k = e^{k \log z}$$

$$k \log z = k \{ \log z + i2\pi l : l \in \mathbb{Z} \}$$

$$= k \log z + i2\pi lk.$$

$$e^{k \log z} = e^{k \log z + i2\pi lk}, \quad l \in \mathbb{Z}$$

$$= e^{k \log z} \cdot e^{i2\pi lk}.$$

$$= e^{k \log z}$$

$$\text{Use the fact } e^{y_1 + y_2} = e^{y_1} \cdot e^{y_2}$$

$$\Rightarrow e^{ky} = e^{y+y+\dots+y} = e^y \dots e^y \quad \text{if } k > 0$$

$$e^{ky} = e^{-y-y-\dots-y} = e^{-y} \dots e^{-y} \quad \text{if } k < 0.$$

$$\text{Use } y = \log(z)$$

$$\therefore e^{k \log z} = \begin{cases} z \times z \times \dots \times z & \text{if } k > 0. \\ z^{-1} \times z^{-1} \times \dots \times z^{-1} & \text{if } k < 0. \end{cases}$$